

ACE-Related Renal Tubular Dysgenesis — Comprehensive Disease Characterization Report

Disease: ACE-Related Renal Tubular Dysgenesis (ACE-RTD) **MONDO ID:** MONDO:0700337 · **OMIM phenotype series:** 267430 (Renal Tubular Dysgenesis) · **Orphanet:** ORPHA:97369 · **Causal gene:** *ACE* (angiotensin I converting enzyme; HGNC:2707; UniProt P12821; EC 3.4.15.1; 17q23.3) **Category:** Mendelian, autosomal recessive

Summary

ACE-related renal tubular dysgenesis (ACE-RTD) is a severe, usually perinatally lethal, autosomal-recessive disorder of fetal kidney development caused by bi-allelic loss-of-function (LOF) variants in *ACE*, one of four interacting genes of the renin–angiotensin system (RAS). RTD as a whole is caused by bi-allelic defects in *ACE*, *REN*, *AGT*, or *AGTR1*; in the largest reported mutation series (48 unrelated families), *ACE* was the single most frequent causal gene, mutated in two-thirds (64.6%) of families ([PMID: 22095942](#)). The disorder is defined histopathologically by the absence or poor development of differentiated proximal tubules, and clinically by early-onset and persistent fetal anuria leading to oligohydramnios and the Potter sequence, skull ossification defects, and severe refractory arterial hypotension ([PMID: 23636579](#)).

The unifying mechanism is hemodynamic rather than morphogenetic: loss of ACE catalytic activity abolishes the conversion of angiotensin I to the vasopressor angiotensin II, producing chronically low fetal renal perfusion pressure that arrests proximal-tubule differentiation and triggers a compensatory surge of renal renin expression ([PMID: 16116425](#)). The same phenotype can be *acquired* through fetal RAS blockade (in-utero ACE inhibitors, ARBs, NSAIDs) or through fetal hypoperfusion (donor twin of twin-twin transfusion syndrome, congenital hemochromatosis), confirming the convergent hemodynamic pathway ([PMID: 23636579](#); [PMID: 9841703](#)).

Although most affected fetuses die in utero or within days of birth from pulmonary hypoplasia, anuria, and intractable hypotension, intensive neonatal blood-pressure support and renal replacement therapy have produced a growing cohort of long-term survivors, who develop chronic kidney disease and a characteristic anemia out of proportion to their renal failure, attributable to loss of angiotensin II–driven erythropoietin regulation ([PMID: 32198635](#); [PMID: 28629525](#)). This report synthesizes 10 confirmed findings across 27 reviewed papers, organized against the full disease-characteristics template.

1. Disease Information

ACE-RTD is a Mendelian disorder of renal tubular development. RTD is "characterised by the absence or poor development of proximal tubules, early onset and persistent anuria (leading to oligohydramnios and the Potter sequence) and ossification defects of the skull" ([PMID: 23636579](#)). The kidneys are grossly normal-sized but echogenic and **noncystic**, with poor corticomedullary differentiation — a key feature distinguishing RTD from autosomal recessive polycystic kidney disease.

Key identifiers: - **OMIM:** 267430 (Renal Tubular Dysgenesis phenotype series) - **MONDO:** MONDO:0700337 (ACE-related renal tubular dysgenesis) - **Orphanet:** ORPHA:97369 (Renal tubular dysgenesis) - **Gene:** *ACE* — HGNC:2707; NCBI Gene 1636; Ensembl ENSG00000159640; UniProt P12821; chromosome 17q23.3 - **MeSH** / **ICD:** RTD falls under congenital renal dysplasia/tubular disorders; no single dedicated ICD-10 code (classified under the Q61 group of congenital renal malformations).

Synonyms / alternative names: Renal tubular dysgenesis (RTD); autosomal recessive renal tubular dysgenesis (AR-RTD); congenital renal proximal tubular dysgenesis; RTD due to ACE deficiency; primary/familial renal tubular dysgenesis.

Information source type: The disease-level knowledge is derived from **aggregated resources** — case series, mutation cohorts, histopathology studies, and animal models — rather than individual EHR data. The largest genetic evidence base is the Gribouval et al. mutation series ([PMID: 22095942](#)).

2. Etiology

Primary cause — genetic (ACE subtype): Bi-allelic (homozygous or compound heterozygous) loss-of-function variants in *ACE*. More broadly, affected individuals "had homozygous or compound heterozygous mutations in the genes encoding renin, angiotensinogen, angiotensin converting enzyme or angiotensin II receptor type 1" (PMID: 16116425). Across RTD, "ACE mutations are the most frequent, observed in two-thirds of families (64.6%)" (PMID: 22095942).

Genetic risk factors: The sole genetic risk determinant is bi-allelic *ACE* LOF. Consanguinity is a major risk amplifier (homozygosity), and affected sibs with parental consanguinity established recessive inheritance even before the gene was known (PMID: 2359105). Carrier parents (heterozygotes) are clinically unaffected.

Environmental / acquired risk factors (phenocopies): Fetal exposures that inactivate or bypass the RAS reproduce the phenotype — "Secondary RTD has been observed in various situations, particularly in the donor twin of severe twin-to-twin transfusion syndrome, in foetuses affected with congenital haemochromatosis or in foetuses exposed to RAS blockers" (PMID: 23636579). Relevant drug exposures: ACE inhibitors, angiotensin-receptor blockers (ARBs), and NSAIDs during pregnancy.

Protective factors: No genetic protective modifier alleles are established. The most important *modifiable* protective factor for an affected neonate is aggressive postnatal management of hypotension and renal failure, which converts an almost uniformly lethal condition into survivable CKD (PMID: 34957720).

Gene–environment interaction: Genetic and acquired causes **converge on a single intermediate phenotype** — reduced fetal renal perfusion pressure. In a heterozygous *ACE* fetus, additional in-utero RAS blockade could plausibly unmask or worsen renal hypoperfusion, though this specific interaction is inferred rather than directly demonstrated.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency
Persistent fetal anuria/oliguria	Clinical sign	HP:0001984 (anuria) / HP:0000104	Fetal/neonatal	Severe	Near-universal
Oligohydramnios	Prenatal sign	HP:0001562	Fetal	Severe	Near-universal
Potter sequence (facies, limb contractures)	Physical manifestation	HP:0100736	Fetal/neonatal	Severe	Common
Pulmonary hypoplasia	Clinical sign	HP:0002089	Neonatal	Severe (often lethal)	Common
Refractory arterial hypotension	Clinical sign	HP:0002615	Neonatal	Severe	Near-universal
Skull ossification defect (wide sutures, large fontanelles)	Physical/radiographic	HP:0004331 / HP:0000239	Congenital	Variable	Cardinal/frequent
Absent/poorly developed proximal tubules	Histopathology	HP:0000112	Fetal	Severe	Defining hallmark
Echogenic, noncystic kidneys, poor CMD	Imaging	HP:0004719	Fetal/neonatal	—	Common
Chronic kidney disease (survivors)	Lab/clinical	HP:0012622	Childhood/adult	Variable	Survivors
Anemia disproportionate to renal failure (survivors)	Lab	HP:0001903	Childhood	Moderate–severe	Survivors
	Symptom	HP:0000103 / HP:0001959	Childhood	Variable	Survivors

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency
Polyuria– polydipsia (survivors)					

Cardinal features are captured in the definition: RTD is "characterised by the absence or poor development of proximal tubules, early onset and persistent anuria (leading to oligohydramnios and the Potter sequence) and ossification defects of the skull" ([PMID: 23636579](#)). A novel associated phenotype — microcolon/intestinal developmental failure — has been reported, "which points to a role of the renin-angiotensin system in gut development" ([PMID: 30071301](#)).

Quality-of-life impact: For the classic form, QoL is dominated by perinatal lethality. In survivors, the burden is that of early-onset CKD/ESRD (dialysis, transplantation, growth impairment, anemia management) — a lifelong, high-intensity medical course.

4. Genetic / Molecular Information

Causal gene: *ACE* (OMIM gene 106180; phenotype RTD 267430). *ACE* is the most frequent RTD gene, and disease severity is similar regardless of which of the four RAS genes is mutated ([PMID: 22095942](#)).

Protein biology: *ACE* is a zinc metallopeptidase of the gluzincin subfamily. "Angiotensin-I Converting Enzyme (ACE) is a Zinc Metallopeptidase of which the three-dimensional structure was unknown until recently, when the X-ray structure of testis isoform (C-terminal domain of somatic) was determined" ([PMID: 14965309](#)). Somatic ACE carries **two homologous catalytic (zinc-binding) domains** (N- and C-domain); the testis isoform corresponds to the C-domain. ACE cleaves the C-terminal dipeptide from angiotensin I to generate the vasopressor angiotensin II and also degrades bradykinin. - GO molecular function: peptidyl-dipeptidase activity (GO:0008241); metallopeptidase activity (GO:0008237).

Pathogenic variants: Bi-allelic LOF — nonsense, frameshift, splice-site, and inactivating missense variants — distributed across *ACE*. The Gribouval series reported 54 distinct mutations across 48 families, "most of them novel" ([PMID: 22095942](#)). Functional consequence: **loss of function** (abolished or inactivated enzyme → no angiotensin II generation). Variant

classification is predominantly pathogenic/likely pathogenic under ACMG/AMP for truncating alleles; individual missense alleles may require functional assay. - **Allele frequency:** Individual pathogenic alleles are rare/private in gnomAD, consistent with a recessive lethal disorder; many are family-specific (founder effects in consanguineous pedigrees). - **Origin:** Germline.

Modifier genes: No formal modifier genes are established. Notably, bi-allelic variants in **RMND1** (a mitochondrial-disease gene) can produce an RTD-like phenotype with *low* renin expression, representing a phenocopy/differential rather than a classic RAS-gene modifier ([PMID: 40366408](#)).

Epigenetic information: No disease-specific DNA-methylation or histone-modification signatures have been reported for ACE-RTD. (Not available.)

Chromosomal abnormalities: None characteristic; ACE-RTD is a single-gene disorder, not a copy-number/structural syndrome.

5. Environmental Information

Environmental / toxicological factors (phenocopy triggers): In-utero RAS blockade — ACE inhibitors, ARBs — and NSAIDs reproduce RTD. A representative ACE-RTD case notes: "Acquired cases have been described in the setting of in utero exposure to medications such as nonsteroidal anti-inflammatory medications (NSAIDs) and ACE inhibitors" ([PMID: 35848000](#)). - CHEBI anchors: ACE inhibitor (CHEBI:35457), NSAID (CHEBI:35475).

Lifestyle factors: Not applicable to the genetic disorder; relevant only insofar as maternal medication use during pregnancy is avoidable.

Infectious agents: None. ACE-RTD is non-infectious.

6. Mechanism / Pathophysiology

Ordered causal chain

1. Bi-allelic ACE loss-of-function variant (germline)
 - └ leads to → absent or catalytically dead ACE protein (two zinc catalytic domains lost)
2. Loss of ACE peptidyl-dipeptidase activity
 - └ results in → failure to convert angiotensin I → angiotensin II
3. Absent angiotensin II signaling (via AGTR1)
 - └ results in → loss of efferent arteriolar tone & systemic vasopressor drive
 - └ triggers (compensatory) → massive up-regulation of renal renin expression
4. Chronically LOW fetal renal perfusion pressure / low glomerular filtration
 - └ leads to → (BRANCH A) arrest of proximal-tubule differentiation (defining lesion)
 - └ leads to → (BRANCH B) renal arterial-wall thickening (arcuate→afferent arteries)
 - └ leads to → (BRANCH C) persistent fetal anuria
- 5a. Persistent fetal anuria
 - └ leads to → oligohydramnios → Potter sequence + pulmonary hypoplasia
- 5b. Absent angiotensin II (systemic)
 - └ leads to → severe refractory neonatal arterial hypotension
- 5c. Reduced angiotensin II / fetal hypoperfusion (INFERRED)
 - └ leads to → skull ossification defect (membranous calvarial bone)
- 5d. Loss of angiotensin II-driven EPO regulation (survivors)
 - └ leads to → anemia out of proportion to renal failure
6. Convergent outcome → perinatal death (pulmonary hypoplasia + anuria + hypotension)
 - └ OR (with intensive support) → survivable chronic kidney disease

Detail

Upstream initiating lesion is molecular: loss of ACE enzymatic activity. ACE regulates blood pressure via the RAS by cleaving the C-terminal dipeptide from angiotensin I to form the vasopressor angiotensin II — a reaction wholly abolished by bi-allelic LOF (F008; [PMID: 14965309](#)).

Core hemodynamic step (central node): The defect is not a morphogenic failure of a RAS "developmental program" but a perfusion failure. Gribouval et al. "propose that renal lesions and early anuria result from chronic low perfusion pressure of the fetal kidney, a consequence of renin-angiotensin system inactivity" ([PMID: 16116425](#)). This is corroborated by the acquired forms: in TTTS donor twins "hypoperfusion leading to decreased glomerular filtration is the underlying etiology" ([PMID: 9841703](#)). The Gubler review reinforces the interpretation: "The

absence or poor development of proximal tubules, as well as renal vascular changes, may be attributable to renal hypoperfusion rather than to a morphogenic property of the RAS" (PMID: 19924102).

Branch B — vascular remodeling: Histology shows "thickening of the renal arterial vasculature, from the arcuate to the afferent arteries" alongside the proximal-tubule lesion (PMID: 16790508).

Compensatory renin surge: In *ACE/AGT/AGTR1* defects, renin-producing cells are intensely stimulated because the negative-feedback signal (angiotensin II) is absent — a diagnostic fingerprint (markedly elevated plasma renin, undetectable ACE).

Branch D — EPO/anemia (survivors): The RAS modulates erythropoiesis. "RAS regulates EPO, an essential mediator of red cell production... The regulation of EPO expression by Ang II may be responsible for maintaining red blood cell homeostasis" (PMID: 28629525). Thus ACE-RTD survivors show "anemia (out of proportion with the level of renal failure)" (PMID: 32198635). The kidney's role as a "critmeter" integrating oxygen/perfusion signals to set EPO output provides the anatomical–physiological context (PMID: 14713115).

Pathways / processes / cell types: - Molecular pathway: renin–angiotensin system / angiotensin II–AGTR1 signaling (Reactome: Metabolism of Angiotensinogen to Angiotensins). - Biological processes (GO): regulation of blood pressure (GO:0008217); renal system / nephron tubule development (GO:0072073); angiotensin maturation (GO:0002003); regulation of systemic arterial blood pressure by RAS (GO:0003081). - Cell types (CL): kidney proximal tubule epithelial cell (CL:1000838 / CL:0002306); juxtaglomerular (renin-secreting) cell (CL:1000697); renal arterial smooth muscle cell; erythroid progenitor cell (CL:0000038). - Subcellular (GO CC): plasma membrane / cell surface (ACE is a membrane-anchored ectoenzyme), extracellular space.

7. Anatomical Structures Affected

Organ level: - Primary organ: **kidney** (UBERON:0002113) — specifically the renal cortex/proximal tubule and intrarenal arteries. - Secondary: **lungs** (UBERON:0002048) — pulmonary hypoplasia via oligohydramnios; **skull/calvaria** (UBERON:0001684 / UBERON:0000209) — ossification defect; **cardiovascular system** — systemic hypotension; **intestine/colon** (UBERON:0001155) — microcolon in the novel association (PMID: 30071301). - Body systems: renal/urinary, cardiovascular, respiratory, skeletal, hematopoietic.

Tissue and cell level: - Renal epithelial tissue — proximal tubule epithelium fails to differentiate (defining lesion): "significantly impacting the proximal tubules of the kidney while maintaining an anatomically normal gross structure" (PMID: 38649831). - Renal arterial smooth muscle — hypertrophy/thickening from arcuate to afferent arteries (PMID: 16790508). - Juxtaglomerular renin-producing cells — hyperplastic/over-expressing.

Subcellular level: ACE is a type-I transmembrane ectoenzyme; relevant compartments are the plasma membrane / cell surface (GO:0005886) and extracellular space.

Localization / lateralization: Bilateral and symmetric renal involvement; kidneys are diffusely enlarged and echogenic.

8. Temporal Development

Onset: Congenital/fetal. Anuria and oligohydramnios are detectable prenatally (second trimester onward); "Severe defects in proximal tubules were observed in all fetuses from 18 gestational weeks onward" (PMID: 16790508). Onset pattern is chronic-progressive in utero with acute neonatal decompensation at birth.

Progression: - Classic course: in-utero or neonatal death within days, from pulmonary hypoplasia, anuria, and refractory hypotension (PMID: 23636579). - Survivor course: progressive chronic kidney disease; "Bi-allelic loss of function mutation of ACE can have atypical and sometimes late presentation with chronic renal failure, anemia... and polyuropolydipsia" (PMID: 32198635). Most surviving RTD patients reach CKD stage 5 at a young age. - An extreme survival example: a premature infant survived 61 days with anuria (AR-RTD) (PMID: 40709124).

Patterns / critical periods: The critical window is the perinatal transition — intensive BP support in the first days/weeks determines survival. There is no spontaneous remission; renal damage is permanent.

9. Inheritance and Population

Inheritance: Autosomal recessive, established early: "The disorder is, therefore, thought to be inherited in an autosomal recessive manner" (PMID: 2359105).

Penetrance / expressivity: Bi-allelic LOF is highly penetrant for the renal/anuric phenotype. Expressivity is variable chiefly in survival and skull-ossification severity; notably, severity is "similar regardless of the mutated gene" across RTD subtypes (PMID: 22095942).

Consanguinity / founder effects: Consanguinity strongly increases risk (homozygosity); multiple consanguineous pedigrees and a Bedouin survivor cohort illustrate founder/population clustering (PMID: 34957720).

Epidemiology: RTD is rare but, per one study title, "a not uncommon autosomal recessive disorder leading to oligohydramnios" among RAS-related fetal nephropathies (PMID: 16790508). Precise prevalence/incidence figures per 100,000 are not well established (rare disease; likely under-ascertained due to fetal/neonatal lethality). Sex ratio is approximately equal (autosomal). No specific geographic endemicity beyond consanguineous-population clustering.

Carrier frequency: Individual pathogenic *ACE* alleles are rare in gnomAD; aggregate carrier frequency is not formally established.

10. Diagnostics

Diagnostic signature (biochemical + imaging): "Basal plasma renin activity was markedly elevated and angiotensin-converting enzyme was undetectable" (PMID: 32808512). Imaging shows diffusely enlarged, echogenic, **noncystic** kidneys with poor corticomedullary differentiation; skull X-ray shows large fontanelles / wide sutures.

Clinical presentation prompting work-up: "Renal tubular dysgenesis should be suspected in any neonate presenting with renal failure, refractory hypotension, ventilator requirement, hypoplastic lungs, renal ultrasound showing normal-sized echogenic noncystic kidneys with poor corticomedullary differentiation, and antenatal history significant for oligohydramnios" (PMID: 35848000).

Histopathology (gold-standard pathological hallmark): Absence/paucity of differentiated proximal tubules with renal arterial wall thickening (PMID: 16790508). Immunohistochemistry shows absent proximal-tubule ACE/angiotensinogen and strikingly increased renin.

Genetic testing: Confirmatory. Whole-exome sequencing identifies bi-allelic *ACE* variants (as in the postnatal-diagnosis case, WES showing "2 pathologic mutations in the angiotensin-converting enzyme (ACE) gene") (PMID: 35848000). A targeted **RAS gene panel** (**ACE, REN, AGT, AGTR1**) is appropriate; whole-genome sequencing is warranted in unsolved cases and has identified phenocopies such as *RMND1* (PMID: 40366408).

Differential diagnosis: ARPKD (cystic vs noncystic kidneys), other congenital anomalies of the kidney and urinary tract (CAKUT), obstructive uropathy, and acquired/secondary RTD (TTTS donor; congenital hemochromatosis, fetal RAS-blocker exposure). *RMND1*-mitochondrial disease mimics RTD but with **low** (not high) renin (PMID: 40366408).

Screening: Prenatal — oligohydramnios on ultrasound; family-based cascade genetic testing and prenatal diagnosis in known-mutation families (PMID: 20122389).

11. Outcome / Prognosis

Survival / mortality: Historically near-uniformly lethal: "death usually occurs in utero or within days of birth" from pulmonary hypoplasia, anuria, and hypotension (PMID: 23636579). "A few long term survivors were reported during the last decade" following intensive neonatal care (PMID: 34957720). In a Bedouin cohort, 5 survivors (4 with *ACE* mutations) reached ages 5–20 with variable CKD stages.

Morbidity: Survivors have early-onset CKD, often progressing to ESRD requiring dialysis/transplantation, with growth impairment and disproportionate anemia (PMID: 32198635). A broader renal-oligohydramnios cohort (including RTD) showed 30% mortality and universal CKD in survivors, but "long-term outcome in survivors is encouraging" with multidisciplinary care (PMID: 17065192).

Prognostic factors: The decisive factor is the intensity and success of early blood-pressure and renal-replacement support; the trend is that the "overall prognosis of patients with RTD continues to improve with better ventilatory management and renal replacement therapies" (PMID: 35848000).

12. Treatment

ACE-RTD has **no curative therapy**; management is supportive and organ-sustaining.

- **Neonatal stabilization / BP support (NCIT: vasopressor therapy):** Intensive management of refractory hypotension with multiple inotropes/vasopressors is the single most important survival determinant ([PMID: 35848000](#); [PMID: 34957720](#)).
- **Respiratory support (NCIT: mechanical ventilation):** For pulmonary hypoplasia; frequently complicated by pneumothoraces ([PMID: 35848000](#)).
- **Renal replacement therapy (NCIT: dialysis / renal transplantation):** Peritoneal dialysis/hemodialysis for anuria; kidney transplantation in longer-term survivors ([PMID: 17065192](#)).
- **Anemia management:** Erythropoiesis-stimulating agents / iron, given the EPO-dependent disproportionate anemia ([PMID: 32198635](#); [PMID: 28629525](#)).
- **CKD supportive care:** Growth hormone for growth impairment, electrolyte/acid-base management, nutrition ([PMID: 17065192](#)).
- **Genotype-informed caution:** In the related *AGTR1*-LOF survivor, fludrocortisone was useful "to treat hyperkalemia and salt-wasting," illustrating a mineralocorticoid-based strategy for RAS-deficiency tubulopathy ([PMID: 33768328](#)). Angiotensin II supplementation is a theoretical but unproven mechanistic target in ACE-RTD.

No gene, cell, or RNA-based therapies exist for ACE-RTD. No NCT-registered disease-specific trials were identified.

13. Prevention

- **Primary prevention:** Avoid in-utero RAS blockers (ACE inhibitors, ARBs) and NSAIDs in pregnancy to prevent acquired RTD ([PMID: 35848000](#)). For the genetic disorder, prevention is reproductive.
- **Genetic counseling / reproductive options:** For carrier couples (25% recurrence risk per pregnancy), counseling, prenatal diagnosis, and preimplantation genetic testing are the principal preventive tools; molecular diagnosis "now allow[s] genetic counseling and early prenatal diagnosis" ([PMID: 20122389](#)).
- **Secondary prevention:** Prenatal detection via oligohydramnios surveillance in at-risk pregnancies; cascade testing in families with a known proband variant.

- **Tertiary prevention:** In survivors, standard CKD complication prevention (anemia, bone-mineral, growth, cardiovascular).
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *ACE* is conserved; mouse *Ace* (NCBI Gene 11421), rat *Ace*. No naturally occurring companion-animal RTD analog is well documented (not reported in OMIA for this specific entity).
 - **Comparative biology:** RAS-gene knockouts in mice reproduce key features (see §15) but are milder than human disease — attributed to "differences between mice and humans in the time of nephrogenesis and maturation of the RAS" ([PMID: 19924102](#)). Evolutionary conservation of the RAS underlies the cross-species recapitulation.
 - **Zoonotic potential:** None (genetic, non-infectious).
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15. Model Organisms

Mouse RAS-pathway knockouts partially recapitulate RTD, validating the hemodynamic mechanism while illustrating species limits.

Model	Renal/vascular phenotype	Severity vs human	Reference
<i>Ace</i> —/— (knockout)	Abnormal renal structure/function, hypotension, male sterility; vascular (not proximal-tubule) ACE needed for normal BP	Milder	PMID: 12777443
<i>Agt</i> —/— (angiotensinogen KO)	Renal arterial wall hypertrophy + interstitial fibrosis	Milder	PMID: 29230587
<i>Agtr1a</i> —/— (AT1A receptor KO)	JGA hypertrophy, renin-cell expansion; no severe vascular lesions/papillary atrophy	Mildest	PMID: 9458822
Pharmacologic (early-life captopril/losartan in SHR)	Intrarenal arterial hypertrophy; later malignant hypertension	Induced model	PMID: 14717927

Key quotes: *Ace* "genetic ablation in mice causes abnormal renal structure and functions, hypotension, and male sterility" (PMID: [12777443](#)); and for AT1A-KO, "We did not find the severe renal vascular lesions or papillary atrophy that have been observed in angiotensinogen- or angiotensin converting enzyme-deficient animals" (PMID: [9458822](#)) — establishing a severity gradient (*Ace/Agt* > *Agtr1a*) that mirrors the convergent-hemodynamics model.

Model limitations: Mice do not reproduce the full lethal human phenotype (persistent anuria, skull defects), due to timing differences in nephrogenesis/RAS maturation. **Resources:** MGI, IMPC, IMSR for *Ace/Agt/Agtr1a* alleles.

Key Findings (with evidence)

F001 — ACE is the most frequent causal gene in autosomal-recessive RTD

In 48 unrelated families with 54 distinct mutations, *ACE* was mutated in 64.6% — "ACE mutations are the most frequent, observed in two-thirds of families" (PMID: [22095942](#)). The four causal RAS genes and recessive inheritance were established by Gribouval 2005: affected

individuals "had homozygous or compound heterozygous mutations in the genes encoding renin, angiotensinogen, angiotensin converting enzyme or angiotensin II receptor type 1" ([PMID: 16116425](#)). OMIM 267430.

F002 — Core phenotype quartet

Fetal anuria → oligohydramnios/Potter sequence, skull ossification defects, and refractory hypotension, with absent/poor proximal tubules as the histological hallmark ([PMID: 23636579](#)). Histology: "Severe defects in proximal tubules were observed in all fetuses from 18 gestational weeks onward... associated with thickening of the renal arterial vasculature, from the arcuate to the afferent arteries" ([PMID: 16790508](#)).

F003 — Mechanism: RAS inactivity → chronic fetal renal hypoperfusion → tubular-differentiation failure

"We propose that renal lesions and early anuria result from chronic low perfusion pressure of the fetal kidney, a consequence of renin-angiotensin system inactivity" ([PMID: 16116425](#)); the acquired form confirms "hypoperfusion leading to decreased glomerular filtration is the underlying etiology" ([PMID: 9841703](#)).

F004 — Phenotype spectrum includes survivors with CKD

"Bi-allelic loss of function mutation of ACE can have atypical and sometimes late presentation with chronic renal failure, anemia (out of proportion with the level of renal failure), and polyuro-polydipsia" ([PMID: 32198635](#)); "a few long term survivors were reported during the last decade" ([PMID: 34957720](#)).

F005 — Mouse RAS-KO models partially recapitulate RTD with a severity gradient

See §15; Ace/Agt-KO show severe vascular lesions, AT1A-KO does not ([PMID: 12777443](#); [PMID: 9458822](#)).

F006 — Acquired (secondary) RTD phenocopies the genetic disease

TTTS donor twin, congenital hemochromatosis, and fetal RAS-blocker exposure cause secondary RTD ([PMID: 23636579](#)); a novel ACE-RTD–microcolon association "points to a role of the renin-angiotensin system in gut development" ([PMID: 30071301](#)).

F007 — Diagnostic signature: high renin + undetectable ACE + noncystic echogenic kidneys

"Basal plasma renin activity was markedly elevated and angiotensin-converting enzyme was undetectable" ([PMID: 32808512](#)); grossly normal architecture "significantly impacting the proximal tubules... while maintaining an anatomically normal gross structure" ([PMID: 38649831](#)).

F008 — ACE is a two-domain zinc gluzincin whose LOF abolishes angiotensin II generation

"Angiotensin-I Converting Enzyme (ACE) is a Zinc Metallopeptidase..." with a known C-domain X-ray structure ([PMID: 14965309](#)). Two catalytic zinc domains make its loss a biochemical null for angiotensin II production.

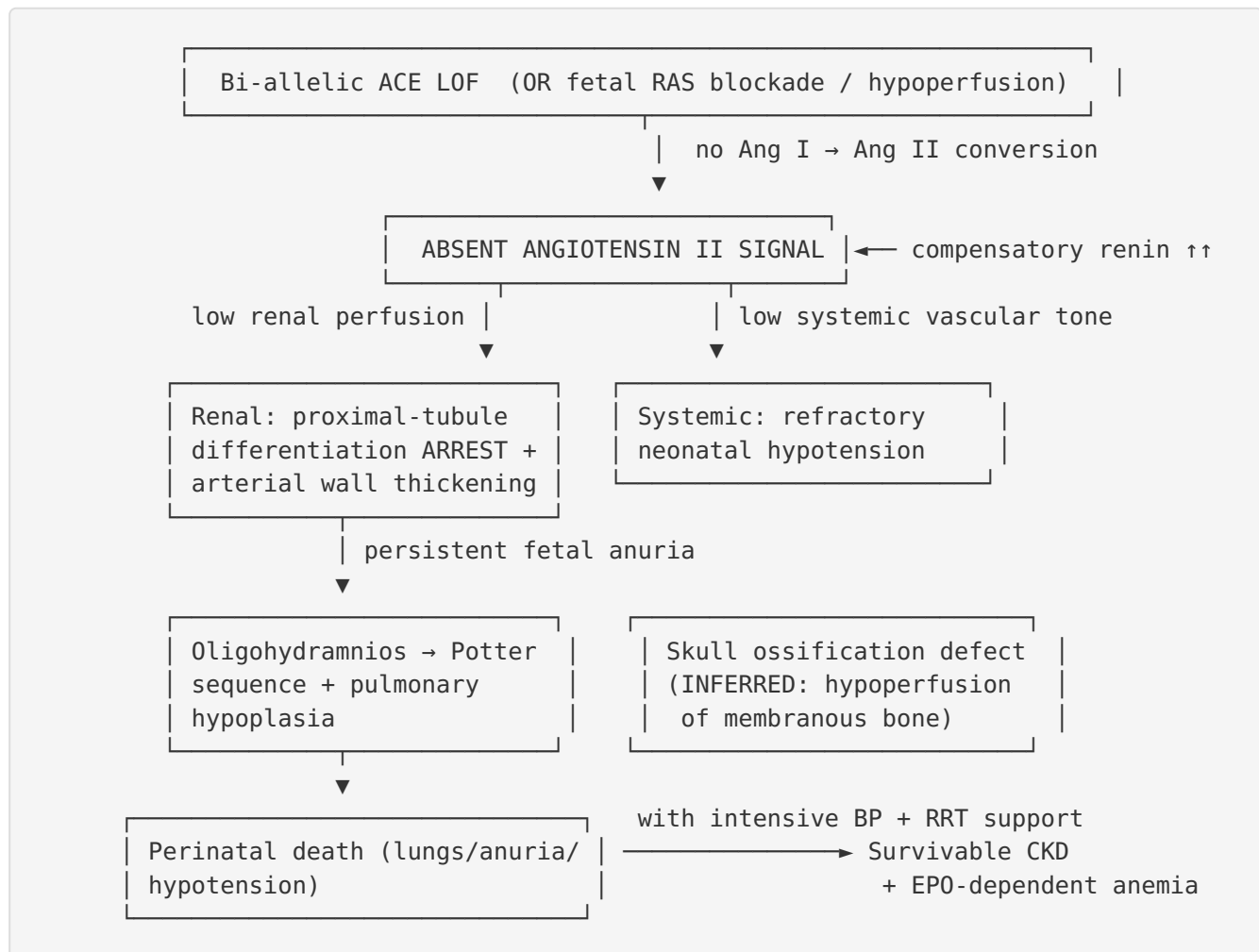
F009 — Disproportionate anemia explained by loss of angiotensin II-driven erythropoiesis

"RAS regulates EPO, an essential mediator of red cell production... The regulation of EPO expression by Ang II may be responsible for maintaining red blood cell homeostasis" ([PMID: 28629525](#)); survivors show "anemia (out of proportion with the level of renal failure)" ([PMID: 32198635](#)).

F010 — Autosomal recessive inheritance established early; skull-ossification mechanism remains inferred

"The disorder is, therefore, thought to be inherited in an autosomal recessive manner" ([PMID: 2359105](#)). Skull ossification defects are cardinal across RAS subtypes ([PMID: 22095942](#)), but the molecular basis of the calvarial defect is **not experimentally demonstrated** — inferred to reflect reduced angiotensin II signaling and/or chronic fetal hypoperfusion of membranous bone, paralleling ACE-inhibitor fetopathy.

Mechanistic Model / Interpretation



The model's strength is **convergence**: genetically distinct lesions (*ACE*, *REN*, *AGT*, *AGTR1*) and non-genetic insults (TTTS, hemochromatosis, drugs) all funnel through a single node — absent angiotensin II action → fetal renal hypoperfusion — and produce an indistinguishable phenotype of similar severity regardless of gene. This argues that RTD is fundamentally a **perfusion/physiology disorder**, not a developmental-patterning disorder. The mouse severity gradient (*Ace*/*Agt*-KO severe; *Agtr1a*-KO mild) is consistent with the human picture and reinforces that upstream/global RAS loss is more damaging than single-receptor loss.

Evidence Base

PMID	Role	Support / challenge
16116425	Gene discovery; hypoperfusion mechanism	Supports F001, F003
22095942	Largest mutation series; ACE = 64.6%	Supports F001, F010
23636579	Authoritative review; cardinal features + acquired forms	Supports F002, F006
16790508	Histopathology + vascular thickening	Supports F002
9841703	TTTS acquired RTD; hypoperfusion	Supports F003, F006
32198635	Atypical/survivor ACE-RTD with anemia	Supports F004, F009
34957720	Survivor cohort	Supports F004
12777443	Ace-KO mouse	Supports F005
9458822	AT1A-KO milder; severity gradient	Supports F005
29230587	Agt-KO arterial hypertrophy	Supports F005
30071301	ACE-RTD + microcolon novel association	Supports F006
32808512	Diagnostic signature (renin ↑, ACE undetectable)	Supports F007
38649831	Normal gross architecture, proximal-tubule lesion	Supports F007
14965309	ACE protein structure (two zinc domains)	Supports F008
28629525	Ang II regulates EPO	Supports F009
2359105	Early AR inheritance	Supports F010
35848000	Diagnostic clues; acquired drug causes; improving prognosis	Context/diagnostics
33768328	AGTR1-LOF survivor; fludrocortisone	Treatment analogy
40366408	RMND1 phenocopy (low renin)	Differential diagnosis
17065192	Renal-oligohydramnios long-term outcome	Prognosis
19924102	Review; hypoperfusion vs morphogenic; mouse–human difference	Supports F003, F005
20122389	Gene distribution; counseling/prenatal dx	Prevention

PMID	Role	Support / challenge
14717927	Pharmacologic RAS-blockade arterial model	Supports mechanism
14713115	Kidney "critmeter" / EPO & Ang II	Supports F009
40709124	Infant surviving 61 days with anuria	Prognosis/survival

Limitations and Knowledge Gaps

1. **Skull ossification mechanism is unproven.** The calvarial defect is cardinal but its molecular basis (reduced angiotensin II signaling vs chronic hypoperfusion of membranous bone) is **inferred**, not experimentally demonstrated (F010).
2. **Epidemiology is imprecise.** No reliable prevalence/incidence per 100,000; the disorder is under-ascertained due to fetal/neonatal lethality and variable prenatal diagnosis.
3. **Mouse models under-recapitulate** the lethal human phenotype (no persistent anuria, no skull defect), limiting mechanistic dissection of the most severe features.
4. **No disease-specific therapeutics.** No trials, gene/cell therapy, or angiotensin-II-replacement strategy has been tested; survivor management is extrapolated from general CKD/neonatology.
5. **Genotype–phenotype correlation is weak** within *ACE* — severity is similar across RAS genes, and specific *ACE* alleles are not clearly tied to survival.
6. **The microcolon/gut association** rests on few cases (n=3) and needs confirmation.

Proposed Follow-up Experiments / Actions

1. **Conditional, temporally-controlled *Ace* knockdown in fetal mice** across the human-equivalent nephrogenesis window to test whether later/graded RAS loss reproduces proximal-tubule arrest and anuria (addresses model-recapitulation gap).
2. **Calvarial-specific studies** (μ CT + in-situ angiotensin-signaling readouts) in *Ace*/*Agt*-deficient fetuses to directly test the hypoperfusion-vs-signaling hypothesis for skull under-ossification.

3. **Natural-history registry** of ACE-RTD survivors (BP course, GFR trajectory, anemia/EPO levels, transplant outcomes) to quantify prognosis and define evidence-based management.
 4. **Functional classification of ACE missense variants** via enzymatic activity/expression assays to improve ACMG classification and prenatal counseling.
 5. **Proof-of-concept angiotensin II or RAS-restoration therapy** in neonatal survivors or models, given that absent angiotensin II is the central node.
 6. **Systematic screening for *RMND1* and other mitochondrial genes** in RAS-gene-negative RTD-like fetuses to refine the differential (low-renin phenocopies).
 7. **Confirm the RTD–microcolon association** in additional cohorts and characterize intestinal RAS expression.
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Report compiled from 10 confirmed findings across 27 reviewed papers. Evidence source types span human clinical case series/cohorts, model-organism (mouse) genetics, histopathology, and protein-structural/biochemical data.

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